2.1 Complex Functions

INTRODUCTION One of the most important concepts in mathematics is that of a function. You may recall from previous courses that a function is a certain kind of correspondence between two sets; more specifically:

> *A **function** f from a set A to a set B is a rule of correspondence that assigns to each element in A one and only one element in B.*

We often think of a function as a rule or a machine that accepts *inputs* from the set A and returns *outputs* in the set B. In elementary calculus we studied functions whose inputs and outputs were real numbers. Such functions are called **real-valued functions of a real variable**. In this section we begin our study of functions whose inputs and outputs are complex numbers. Naturally, we call these functions **complex functions of a complex variable**, or **complex functions** for short. As we will see, many interesting and useful complex functions are simply generalizations of well-known functions from calculus.

A review of function terminology can be found in Section 2.1 of the student study guide.

Function Suppose that f is a function from the set A to the set B. If f assigns to the element a in A the element b in B, then we say that b is the **image** of a under f, or the **value** of f at a, and we write $b = f(a)$. The set A—the set of inputs—is called the **domain** of f and the set of images in B—the set of outputs—is called the **range** of f. We denote the domain and range of a function f by $\text{Dom}(f)$ and $\text{Range}(f)$, respectively. As an example, consider the "squaring" function $f(x) = x^2$ defined for the real variable x. Since any real number can be squared, $\text{Dom}(f)$ is the set $\mathbf{R}$ of all real numbers. The range of f consists of all real numbers x^2 where x is a real number. Of course, $x^2 \geq 0$ for all real x, and it is easy to see from the graph of f that the range of f is the set of all *nonnegative* real numbers. Thus, $\text{Range}(f)$ is the interval $[0, \infty)$. The range of f need not be the same as the set B. For instance, f can be viewed as a function from $A = \mathbf{R}$ to $B = \mathbf{R}$, in which case the range is contained in but not equal to B.

As the following definition indicates, a *complex* function is a function whose inputs and outputs are complex numbers.

DEFINITION 2.1.1 Complex Function

A **complex function** is a function f whose domain and range are subsets of the set $\mathbf{C}$ of complex numbers.

Notation used throughout this text.

A complex function is also called a **complex-valued function of a complex variable**. For the most part we will use the usual symbols f, g, and h to denote complex functions. In addition, inputs to a complex function f will typically be denoted by the variable z and outputs by the variable $w = f(z)$. When referring to a complex function we will use three notations interchangeably, for example, $f(z) = z - i$, $w = z - i$, or, simply, the function $z - i$. Throughout this text the notation $w = f(z)$ will always denote a complex function, whereas the notation $y = f(x)$ will be reserved to represent a real-valued function of a real variable x.

EXAMPLE 1 Complex Function

(**a**) The expression $z^2-(2+i)z$ can be evaluated at any complex number z and always yields a single complex number, and so $f(z) = z^2 - (2 + i)z$ defines a complex function. Values of f are found by using the arithmetic operations for complex numbers given in Section 1.1. For instance, at the points $z = i$ and $z = 1 + i$ we have:

$$f(i) = (i)^2 - (2 + i)(i) = -1 - 2i + 1 = -2i$$

and $$f(1 + i) = (1 + i)^2 - (2 + i)(1 + i) = 2i - 1 - 3i = -1 - i.$$

(**b**) The expression $g(z) = z+2\,\mathrm{Re}\,(z)$ also defines a complex function. Some values of g are:

$$g(i) = i + 2\,\mathrm{Re}\,(i) = i + 2(0) = i$$

and $$g(2 - 3i) = 2 - 3i + 2\,\mathrm{Re}\,(2 - 3i) = 2 - 3i + 2(2) = 6 - 3i.$$ ☰

When the domain of a complex function is not explicitly stated, we assume the domain to be the set of all complex numbers z for which $f(z)$ is defined. This set is sometimes referred to as the *natural domain* of f. For example, the functions $f(z) = z^2 - (2 + i)z$ and $g(z) = z + 2\,\mathrm{Re}(z)$ in Example 1 are defined for all complex numbers z, and so, $\mathrm{Dom}(f) = \mathbf{C}$ and $\mathrm{Dom}(g) = \mathbf{C}$. The complex function $h(z) = z/\left(z^2 + 1\right)$ is not defined at $z = i$ and $z = -i$ because the denominator z^2+1 is equal to 0 when $z = \pm i$. Therefore, $\mathrm{Dom}(h)$ is the set of all complex numbers except i and $-i$.

In the section introduction, we defined a real-valued function of a real variable to be a function whose domain and range are subsets of the set **R** of real numbers. Because **R** is a subset of the set **C** of the complex numbers, *every* real-valued function of a real variable is also a complex function. We will soon see that real-valued functions of two real variables x and y play an important role in the study of complex analysis. In order to avoid repeating the cumbersome terminology *real-valued function of a real variable* and *real-valued function of two real variables,* we use the term **real function** from this point on to refer to any type of function studied in a single or multivariable calculus course.

Real and Imaginary Parts of a Complex Function It is often helpful to express the inputs and the outputs of a complex function in terms of their real and imaginary parts. If $w = f(z)$ is a complex function, then the image of a complex number $z = x+iy$ under f is a complex number $w = u+iv$. By simplifying the expression $f(x + iy)$, we can write the real variables u and v in terms of the real variables x and y. For example, by replacing the symbol z with $x + iy$ in the complex function $w = z^2$, we obtain:

$$w = u + iv = (x + iy)^2 = x^2 - y^2 + 2xyi. \tag{1}$$

From (1) the real variables u and v are given by $u = x^2-y^2$ and $v = 2xy$, respectively. This example illustrates that, if $w = u + iv = f(x + iy)$ is a complex function, then both u and v are real functions of the two real variables x and y. That is, by setting $z = x + iy$, we can express any complex function $w = f(z)$ in terms of two real functions as:

$$f(z) = u(x, y) + iv(x, y). \tag{2}$$

The functions $u(x, y)$ and $v(x, y)$ in (2) are called the **real** and **imaginary parts of** f, respectively.

EXAMPLE 2 Real and Imaginary Parts of a Function

Find the real and imaginary parts of the functions: **(a)** $f(z) = z^2 - (2 + i)z$ and **(b)** $g(z) = z + 2\,\mathrm{Re}(z)$.

Solution In each case, we replace the symbol z by $x + iy$, then simplify.

(a) $f(z) = (x + iy)^2 - (2 + i)(x + iy) = x^2 - 2x + y - y^2 + (2xy - x - 2y)i$. So, $u(x, y) = x^2 - 2x + y - y^2$ and $v(x, y) = 2xy - x - 2y$.

(b) Since $g(z) = x + iy + 2\,\mathrm{Re}(x + iy) = 3x + iy$, we have $u(x, y) = 3x$ and $v(x, y) = y$. ☰

A function f can be defined without using the symbol z.

Every complex function is completely determined by the real functions $u(x, y)$ and $v(x, y)$ in (2). Thus, a complex function $w = f(z)$ can be defined by arbitrarily specifying two real functions $u(x, y)$ and $v(x, y)$, even though $w = u + iv$ may not be obtainable through familiar operations performed solely on the symbol z. For example, if we take, say, $u(x, y) = xy^2$ and $v(x, y) = x^2 - 4y^3$, then $f(z) = xy^2 + i(x^2 - 4y^3)$ defines a complex function. In order to find the value of f at the point $z = 3 + 2i$, we substitute $x = 3$ and $y = 2$ into the expression for f to obtain $f(3 + 2i) = 3 \cdot 2^2 + i(3^2 - 4 \cdot 2^3) = 12 - 23i$.

We note in passing that complex functions defined in terms of $u(x, y)$ and $v(x, y)$ can always be expressed, if desired, in terms of operations on the symbols z and $\bar{z}$. See Problem 32 in Exercises 2.1.

Exponential Function In Section 1.6 we informally introduced the complex exponential function e^z. This complex function is an example of one that is defined by specifying its real and imaginary parts.

DEFINITION 2.1.2 Complex Exponential Function

The function e^z defined by:

$$e^z = e^x \cos y + ie^x \sin y \tag{3}$$

is called the **complex exponential function**.

By Definition 2.1.2, the real and imaginary parts of the complex exponential function are $u(x, y) = e^x \cos y$ and $v(x, y) = e^x \sin y$, respectively. Values of the complex exponential function $w = e^z$ are found by expressing the point z as $z = x + iy$ and then substituting the values of x and y in (3). The following example illustrates this procedure.

EXAMPLE 3 Values of the Complex Exponential Function

Find the values of the complex exponential function e^z at the following points.

(a) $z = 0$ **(b)** $z = i$ **(c)** $z = 2 + \pi i$

Solution In each part we substitute $x = \mathrm{Re}(z)$ and $y = \mathrm{Im}(z)$ into (3) and then simplify.

(**a**) For $z = 0$, we have $x = 0$ and $y = 0$, and so $e^0 = e^0 \cos 0 + ie^0 \sin 0$. Since $e^0 = 1$ (for the real exponential function), $\cos 0 = 1$, and $\sin 0 = 0$, $e^0 = e^0 \cos 0 + i \sin 0$ simplifies to $e^0 = 1$.

(**b**) For $z = i$, we have $x = 0$ and $y = 1$, and so:

$$e^i = e^0 \cos 1 + ie^0 \sin 1 = \cos 1 + i \sin 1 \approx 0.5403 + 0.8415i.$$

(**c**) For $z = 2 + \pi i$, we have $x = 2$ and $y = \pi$, and so $e^{2+\pi i} = e^2 \cos \pi + ie^2 \sin \pi$. Since $\cos \pi = -1$ and $\sin \pi = 0$, this simplifies to $e^{2+\pi i} = -e^2$. ☰

Exponential Form of a Complex Number The complex exponential function enables us to express the polar form of a nonzero complex number $z = r(\cos \theta + i \sin \theta)$ in a particularly convenient and compact form:

$$z = re^{i\theta}. \tag{4}$$

We call (4) the **exponential form** of the complex number z. For example, a polar form of the complex number $3i$ is $3\,[\cos(\pi/2) + i \sin(\pi/2)]$, whereas an exponential form of $3i$ is $3e^{i\pi/2}$. Bear in mind that in the exponential form (4) of a complex number, the value of $\theta = \arg(z)$ is not unique. This is similar to the situation with the polar form of a complex number. You are encouraged to verify that $\sqrt{2}e^{i\pi/4}$, $\sqrt{2}e^{i9\pi/4}$, and $\sqrt{2}e^{i17\pi/4}$ are all valid exponential forms of the complex number $1 + i$.

If z is a real number, that is, if $z = x + 0i$, then (3) gives $e^z = e^x \cos 0 + ie^x \sin 0 = e^x$. In other words, the complex exponential function agrees with the usual real exponential function for real z. Many well-known properties of the real exponential function are also satisfied by the complex exponential function. For instance, if z_1 and z_2 are complex numbers, then (3) can be used to show that:

$$e^0 = 1, \tag{5}$$

$$e^{z_1} e^{z_2} = e^{z_1 + z_2}, \tag{6}$$

$$\frac{e^{z_1}}{e^{z_2}} = e^{z_1 - z_2}, \tag{7}$$

$$\left(e^{z_1}\right)^n = e^{nz_1} \text{ for } n = 0, \pm 1, \pm 2, \ldots. \tag{8}$$

Proofs of properties (5)–(8) will be given in Section 4.1 where the complex exponential function is discussed in greater detail.

While the real and complex exponential functions have many similarities, they also have some surprising and important differences. Perhaps the most unexpected difference is:

• *The complex exponential function is periodic.*

In Problem 33 in Exercises 2.1 you are asked to show that the complex exponential function has pure imaginary period $2\pi i$.

Polar Coordinates Up to this point, the real and imaginary parts of a complex function were determined using the Cartesian description $x + iy$ of the complex variable z. It is equally valid, and, oftentimes, more convenient to express the complex variable z using either the polar form $z = r(\cos \theta + i \sin \theta)$ or, equivalently, the exponential form $z = re^{i\theta}$. Given a complex function $w = f(z)$, if we replace the symbol z with $r(\cos \theta + i \sin \theta)$, then we can write this function as:

$$f(z) = u(r, \theta) + iv(r, \theta). \tag{9}$$

We still call the real functions $u(r,\ \theta)$ and $v(r,\ \theta)$ in (9) the real and imaginary parts of f, respectively. For example, replacing z with $r(\cos\theta + i\sin\theta)$ in the function $f(z) = z^2$, yields, by de Moivre's formula,

$$f(z) = (r(\cos\theta + i\sin\theta))^2 = r^2\cos 2\theta + ir^2\sin 2\theta.$$

Thus, using the polar form of z we have shown that the real and imaginary parts of $f(z) = z^2$ are

The functions $u(r, \theta)$ and $v(r, \theta)$ are not the same as the functions $u(x,\ y)$ and $v(x,\ y)$.

$$u(r, \theta) = r^2\cos 2\theta \quad \text{and} \quad v(r, \theta) = r^2\sin 2\theta, \tag{10}$$

respectively. Because we used a polar rather than a Cartesian description of the variable z, the functions u and v in (10) are not the same as the functions u and v in (1) previously computed for the function z^2.

As with Cartesian coordinates, a complex function can be defined by specifying its real and imaginary parts in polar coordinates. The expression $f(z) = r^3\cos\theta + (2r\sin\theta)i$, therefore, defines a complex function. To find the value of this function at, say, the point $z = 2i$, we first express $2i$ in polar form:

$$2i = 2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right).$$

We then set $r = 2$ and $\theta = \pi/2$ in the expression for f to obtain:

$$f(2i) = (2)^3\cos\frac{\pi}{2} + \left(2\cdot 2\sin\frac{\pi}{2}\right)i = 8\cdot 0 + (4\cdot 1)i = 4i.$$

Remarks COMPARISON WITH REAL ANALYSIS

(*i*) The complex exponential function provides a good example of how complex functions can be similar to and, at the same time, different from their real counterparts. Both the complex and the real exponential function satisfy properties (5)–(8). On the other hand, the complex exponential function is periodic and, from part (c) of Example 3, a value of the complex exponential function can be a negative real number. Neither of these properties are shared by the real exponential function.

(*ii*) In this section we made the important observation that every complex function can be defined in terms of two real functions $u(x,\ y)$ and $v(x,\ y)$ as $f(z) = u(x,\ y) + iv(x,\ y)$. This implies that the study of complex functions is closely related to the study of real multivariable functions of two real variables. The notions of limit, continuity, derivative, and integral of such real functions will be used to develop and aid our understanding of the analogous concepts for complex functions.

(*iii*) On page 48 we discussed that real-valued functions of a real variable and real-valued functions of two real variables can be viewed as special types of complex functions. Other special types of complex functions that we encounter in the study of complex analysis include the following:

Real-valued functions of a complex variable are functions $y = f(z)$ where z is a complex number and y is a real number. The functions $x = \text{Re}(z)$ and $r = |z|$ are both examples of this type of function.

Complex-valued functions of a real variable are functions $w = f(t)$ where t is a real number and w is a complex number. It is customary to express such functions in terms of two real-valued functions of the real variable t, $w(t) = x(t) + iy(t)$. For example, $w(t) = 3t + i\cos t$ is a complex-valued function of a real variable t.

These special types of complex functions will appear in various places throughout the text.

EXERCISES 2.1

Answers to selected odd-numbered problems begin on page ANS-6.

In Problems 1–8, evaluate the given complex function f at the indicated points.

1. $f(z) = z^2\bar{z} - 2i$	**(a)** $2i$	**(b)** $1 + i$	**(c)** $3 - 2i$
2. $f(z) = -z^3 + 2z + \bar{z}$	**(a)** i	**(b)** $2 - i$	**(c)** $1 + 2i$
3. $f(z) = \log_e \lvert z\rvert + i\,\mathrm{Arg}(z)$	**(a)** 1	**(b)** $4i$	**(c)** $1 + i$
4. $f(z) = \lvert z\rvert^2 - 2\mathrm{Re}(iz) + z$	**(a)** $3 - 4i$	**(b)** $2 - i$	**(c)** $1 + 2i$
5. $f(z) = (xy - x^2) + i(3x + y)$	**(a)** $3i$	**(b)** $4 + i$	**(c)** $3 - 5i$
6. $f(z) = e^z$	**(a)** $2 - \pi i$	**(b)** $\frac{\pi}{3}i$	**(c)** $\log_e 2 - \frac{5\pi}{6}i$
7. $f(z) = r + i\cos^2\theta$	**(a)** 3	**(b)** $-2i$	**(c)** $2 - i$
8. $f(z) = r\sin 3\theta + i\cos 2\theta$	**(a)** -2	**(b)** $1 + i$	**(c)** $-5i$

In Problems 9–16, find the real and imaginary parts u and v of the given complex function f as functions of x and y.

9. $f(z) = 6z - 5 + 9i$

10. $f(z) = -3z + 2\bar{z} - i$

11. $f(z) = z^3 - 2z + 6$

12. $f(z) = z^2 + \bar{z}^2$

13. $f(z) = \dfrac{\bar{z}}{z + 1}$

14. $f(z) = z + \dfrac{1}{z}$

15. $f(z) = e^{2z+i}$

16. $f(z) = e^{z^2}$

In Problems 17–22, find the real and imaginary parts u and v of the given complex function f as functions of r and θ.

17. $f(z) = \bar{z}$

18. $f(z) = \lvert z\rvert$

19. $f(z) = z^4$

20. $f(z) = z + \dfrac{1}{z}$

21. $f(z) = e^z$

22. $f(z) = x^2 + y^2 - yi$

In Problems 23–26, find the natural domain of the given complex function f.

23. $f(z) = 2\mathrm{Re}(z) - iz^2$

24. $f(z) = \dfrac{3z + 2i}{z^3 + 4z^2 + z}$

25. $f(z) = \dfrac{iz}{\lvert z - 1\rvert}$

26. $f(z) = \dfrac{iz}{\lvert z\rvert - 1}$

Focus on Concepts

27. Discuss: Do the following expressions define complex functions $f(z)$? Defend your answer.

(a) $\arg(z)$ **(b)** $\mathrm{Arg}(z)$ **(c)** $\cos(\arg(z)) + i\sin(\arg(z))$

(d) $z^{1/2}$ **(e)** $\lvert z\rvert$ **(f)** $\mathrm{Re}(z)$

28. Find the range of each of the following complex functions.

(**a**) $f(z) = \text{Im}(z)$ defined on the closed disk $|z| \le 2$

(**b**) $f(z) = (1+i)\,|z|$ defined on the square $0 \le \text{Re}(z) \le 1, 0 \le \text{Im}(z) \le 1$

(**c**) $f(z) = \bar{z}$ defined on the upper half-plane $\text{Im}(z) > 0$

29. Find the natural domain and the range of each of the following complex functions.

(**a**) $f(z) = \dfrac{z}{|z|}$. [*Hint*: In order to determine the range, consider $|f(z)|$.]

(**b**) $f(z) = 3 + 4i + \dfrac{5z}{|z|}$.

(**c**) $f(z) = \dfrac{z+\bar{z}}{z-\bar{z}}$.

30. Give an example of a complex function whose natural domain consists of all complex numbers except 0, $1+i$, and $1-i$.

31. Determine the natural domain and range of the complex function $f(z) = \cos(x-y) + i\sin(x-y)$.

32. Suppose that $z = x + iy$. Reread Section 1.1 and determine how to express x and y in terms of z and $\bar{z}$. Then write the following functions in terms of the symbols z and $\bar{z}$.

(**a**) $f(z) = x^2 + y^2$ (**b**) $f(z) = x - 2y + 2 + (6x+y)i$

(**c**) $f(z) = x^2 - y^2 - (5xy)i$ (**d**) $f(z) = 3y^2 + \left(3x^2\right)i$

33. In this problem we examine some properties of the complex exponential function.

(**a**) If $z = x + iy$, then show that $|e^z| = e^x$.

(**b**) Are there any complex numbers z with the property that $e^z = 0$? [*Hint*: Use part (a).]

(**c**) Show that $f(z) = e^z$ is a function that is periodic with pure imaginary period $2\pi i$. That is, show that $e^{z+2\pi i} = e^z$ for all complex numbers z.

34. Use (3) to prove that $e^{\bar{z}} = \overline{e^z}$ for all complex z.

35. What can be said about z if $\left|e^{-z}\right| < 1$?

36. Let $f(z) = \dfrac{e^{iz} + e^{-iz}}{2}$.

(**a**) Show that f is periodic with real period 2π.

(**b**) Suppose that z is real. That is, $z = x + 0i$. Use (3) to rewrite $f(x+0i)$. What well-known real function do you get?

37. What is the period of each of the following complex functions?

(**a**) $f(z) = e^{z+\pi}$ (**b**) $f(z) = e^{\pi z}$

(**c**) $f(z) = e^{2iz}$ (**d**) $f(z) = e^{3z+i}$

38. If $f(z)$ is a complex function with pure imaginary period i, then what is the period of the function $g(z) = f(iz-2)$?

Computer Lab Assignments

In Problems 39–42, use a CAS to evaluate the given complex function at the indicated points. In *Mathematica* you can define a function using the syntax **f[z_]:=** *an expression in the variable* z. To evaluate the function at the point $a+bi$, use the syntax **f[a+b*I]**.

39. the function and points in Problem 1.

40. the function and points in Problem 2.

41. the function and points in Problem 3.

42. the function and points in Problem 4.